

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (S, FE) / S6 (PT) (S) Examination January 2024 (2019 Scheme)

Course Code: CST302**Course Name: COMPILER DESIGN**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- 1 Describe input buffering scheme in lexical analyser. (3)
- 2 Find regular definition for tokens in the generated strings of the following grammar. (3)
$$S \rightarrow \text{if } E \text{ then } S \mid \text{if } E \text{ then } S \text{ else } S \mid \epsilon$$
$$E \rightarrow T \text{ relop } T \mid T$$
$$T \rightarrow \text{id} \mid \text{num}$$
- 3 What is left recursion? Explain the rule to eliminate left recursion. (3)
- 4 Describe sets-of-items construction algorithm for SLR parser. (3)
- 5 Explain synthesized attributes and inherited attributes. (3)
- 6 Check whether the given translation scheme is L-attributed or not? Justify (3)
$$A \rightarrow B \{ B.i = C.s \} C \{ C.i = A.i \}$$
- 7 What are the advantages of indirect triple compared to triple. (3)
- 8 Write three-address code for a **while** loop by choosing a suitable example. (3)
- 9 Differentiate local and global optimizations. (3)
- 10 Illustrate the role of register descriptor and address descriptor in code generation phase. (3)

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

- 11 a) Draw transition diagram for designing lexical analyser for the tokens such as arithmetical operator, relational operator, identifier and unsigned number. (7)
- b) Write code for the lexical analyser for the above design. (7)

OR

- 12 a) Explain LEX program structure with a sample program. (7)
 b) Explain YACC program structure with a sample program. (7)

Module II

- 13 a) Illustrate design of recursive descent parser with a suitable grammar. (7)
 b) Explain any two drawbacks of recursive descent parser and its solutions. (7)

OR

- 14 a) Write and explain algorithm for constructing LL(1) predictive parsing table. (7)
 b) Construct predictive parsing table for given grammar. (7)

$S \rightarrow SAaAb \mid SBbBa \mid \epsilon$
 $A \rightarrow t$
 $B \rightarrow c$

Module III

- 15 a) Illustrate actions according to operator precedence parser for the input $id_1 + id_2 * id_3$ based on the given grammar. (7)

$E \rightarrow E + E \mid E * E \mid (E) \mid id$

- b) Explain how to construct an operator grammar without ambiguity. (7)

OR

- 16 Construct Canonical LR (1) parsing table and perform parsing actions for a valid input according to given grammar. (14)

$S \rightarrow Aa \mid bAc \mid cAb$
 $A \rightarrow a$

Module IV

- 17 a) Write syntax-directed definition for simple type declaration involving basic types such as int, float and char (assume syntax for C-programming language). (7)
 b) Write SDD for generating syntax tree for arithmetic operations. (7)

OR

- 18 a) Draw DAG representation for the given statement. (5)

$s = (a + b) * (b + c) + (a + b)$

- b) Construct quadruple, triple and indirect triple tables for above DAG representation. (9)

Module V

- 19 a) Explain any three code optimization transformation. (6)
 b) Perform common sub-expression elimination for the following three-address code and represent it as a quadruple table. (8)

```
t1 = a + b
x  = t1
t2 = a + b
t3 = t2 + c
b  = t2
t4 = a + b
y  = t4
```

OR

- 20 a) Neatly explain code generation algorithm and *getreg* function. (7)
- b) Convert to three-address code and write machine code for given statement. (7)

$x = a / b + a / b * (c-d)$
